
Leakage-Aware Optional-Modality Fusion for Alzheimer’s Biomarker Classification from MRI and PET

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GitHub Repository: github.com/mahadamir19/Deep-Learning-Project

Abstract

Prediction of Alzheimer’s disease (AD) from neuroimages using deep learning presents itself as an important but challenging task because MRI and PET neuroimages offer complementary structural-functional information together with acquisition-specific nuisances. In the original survey of the state of the art, our project suggested disease-acquisition latent factorization for domain-invariant learning on multimodal MRI-PET images. As the project unfolded, its scale was adjusted to a feasible level for students’ work: class-imbalance-aware, leakage-aware, and cross-cohort-stable multimodal classification with handling of missing data and PET modality. In this paper, we present our full project progress from the initial baseline model to first improvement and, finally, to second improvement. The baseline model employed a cross-transformer-style multimodal MRI-PET classifier that showed reasonable raw accuracy yet collapsed to the class majority leading to zero sensitivity and zero F1 of the positive class. First improvement introduced encoding of MRI/PET separately, cross-modal fusion, harmonization and prototype reliability approaches, and missing data handling while still being unstable for the very small ds006148 paired datasets. Second improvement included a more grounded design with MRI anchoring and optional use of PET, multi-scale fusion, balanced training, threshold validation, and reliability score calculation. Finally, in the best controlled split of subjects, the improved model showed 82.54%, 63.73%, and 73.69% accuracy, balanced accuracy, and AUC correspondingly.

1. Introduction

Classification of Alzheimer’s disease by means of neuroimaging can be considered as identifying a disease or biomarker class based on brain images. Here, MRI gives structural characteristics, whereas PET gives func-

tional/molecular ones. Therefore, a multimodal solution allows for taking advantage of complementary evidence provided by the two modalities.

In the end, the main issue is that medical imaging datasets come from different acquisition processes (across different sites, MRI scanners and field strengths, reconstruction pipelines, tracers, etc.). Thus, the deep neural network may exploit the shortcuts rather than extracting meaningful disease signal. Consequently, there are two dangers: overestimating internal metrics and failing external test. Therefore, our state-of-the-art review was motivated by a more general question: can representations from MRI-PET scans be factored into acquisition- and disease-related features?

When we came to the implementation part, the scope was narrowed to actually implementing and testing an approach that can be applied to the dataset available in the current environment. The following are the goals of the final project:

- Data splitting must be subject-wise to minimize any potential leakage.
- The model should avoid the majority class collapse problem identified with the baseline.
- The solution should cope with real modality issues, such as PET being optional and less reliable.

Contributions of this work include the following:

- Implementation and evaluation of leakage-aware multimodal MRI-PET baseline with the identified failure mode: high specificity, yet no sensitivity in the positive class.
- Intermediate steps with cross-modal fusing, various harmonization attempts, and prototypes for reliability.
- Final practical grounded system with optional modality fusing, multi-level representation, balancing, thresholding, reliability scoring.
- Reporting not only the accuracy but also balanced accuracy, AUC, sensitivity, specificity, F1, MCC, and confusion matrix because accuracy masks class imbalance problems.

2. Related Work

2.1. Multimodal Alzheimer’s prediction

With recent progress in AD deep learning, multimodal fusion of MRI and PET modalities became increasingly common. Since MRI and PET capture distinct biological processes in the brain, multimodal fusion is an interesting problem. Recent work on multimodal dementia classification with transformers include DiaMond (Li et al., 2025). More broadly, surveys like (Guan et al., 2025) categorize multimodal neuroimaging methods based on fusion type, availability of modalities, and clinical applications. In addition to multimodality, the importance of self-supervised learning increased due to scarcity of labeled data in neuroimaging. Ali et al. introduced self-supervised multimodal alignment based on MRI-PET registration, cross-temporal consistency, and invariant representation across sites (Ali et al., 2025). All these works motivated us to consider two encoders and fusion for classification.

2.2. Domain shift, harmonization, and acquisition nuisance

The next relevant line of research studies the effect of acquisition nuisance on AD prediction by comparing test and training sets on cohort level or across image acquisitions. Fu et al. introduce domain adaptation approach that employs multi-modal CNN feature encoders, multi-head attention-based multimodal fusion, and adversarial domain transfer module (Fu et al., 2025). Another interesting work considers disease-driven domain generalization that emphasizes disease-related features transferability (Lteif et al., 2024). Image-level harmonization techniques like amyloid PET harmonization (Choi et al., 2024) and HACA3 for MRI (Zuo et al., 2023) indicate that acquisition nuisances can be mitigated either prior to or along with machine learning. Our gap in literature is that these methods rely on global feature alignment/harmonization but do not explicitly demonstrate separation between disease and acquisition-related signals.

2.3. Evaluation rigor and updated project scope

Another important take away from existing literature is that unreliable evaluation may mislead about effectiveness of AD deep learning approaches. According to Young et al., data leakage and inadequate validation lead to biased AD classification reports (Young et al., 2025). It is important because our approach involves subject-wise splitting, which makes results more reliable. The initial scope of the SOA project aimed at full latent factorization of disease and acquisition and residual leakage probe. However, in practice the scope shrank due to limitation with paired data and computational constraints. So, the final system is a step toward achieving SOA objectives.

3. Methodology

3.1. Problem formulation

The data will consist of MRI volume x^m , PET volume x^p , modality-availability mask $a \in \{0, 1\}^2$, and binary label $y \in \{0, 1\}$. In the end, the goal will be binary classification, where $y = 1$ corresponds to positive disease/biomarker class and $y = 0$ to the negative/class control. Model f_θ predicts the class logits $z = f_\theta(x^m, x^p, a)$ and class probabilities $\hat{p} = \text{softmax}(z)$. The key is to find a representation which will remain diagnostic but not collapse into the majority class.

3.2. Data and leakage-aware splitting

MRI and PET neuroimages were taken with subject IDs and labels from the ds006148 project notebooks. In the final controlled split, there were 156 samples from 52 subjects. Scans of one subject remained in the same set, making the split subject-wise. In the final run, splits’ train/validation/test sizes were 69/24/63 samples. Splitting by subject was chosen, because split-by-scans would cause subject leakage and performance inflation.

The broader scope of experiments based on ds006148 data was also considered, due to its importance for domain generalization. However, the experiments showed the limited paired dataset of ds006148. As such, the controlled subject-wise splitting of ds006148 is used as the base of claims, but extra results on ds006148 are provided to illustrate further limitations.

3.3. Baseline: corrected multimodal cross-transformer

The baseline notebook proposed a corrected multimodal MRI-PET classifier inspired by multimodal cross-attentive fusion and baselines from literature. MRI and PET images were separately encoded using 3D encoders. Then, embeddings were fused using cross-attentive (transformer-based) fusion block before being processed by a classification head. This baseline was helpful, because it could show how good is the direct MRI-PET fusion by itself.

As the baseline result shows, accuracy alone does not provide enough information. Even though raw accuracy is about 75%, this model fails to predict any positive class samples and predicts only the majority class. All its sensitivity, F1 score, and MCC score are 0, while specificity is 100%. This means, that the baseline only learns a safe negative prediction policy.

3.4. First improvement: multimodal reliability and harmonization

The first improvement was an introduction of explicit multimodal reliability through the use of dual 3D MRI/PET en-

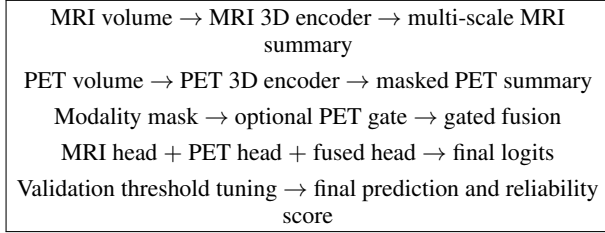


Figure 1. Final second-improvement architecture. The design is deliberately simple: MRI remains the stable anchor, PET contributes only through a learned optional gate, and the decision threshold is tuned on validation metrics rather than fixed at 0.5.

coders, cross-modal fusion, a harmonization-inspired block, and prototype-based distance-based OOD detection.

This conceptual step helped to get closer to the SOA principle of handling variations during acquisition rather than concatenating MRI and PET modalities.

However, this improvement also revealed a practical limitation. When trained in the ds006148 scenario, especially with limited number of paired MRI/PET examples, the network became too big for the amount of data. That’s why the next improvement was not an addition of more complicated blocks, but rather a simplification and more practical implementation of the idea, based on class balance, missing-modality tolerance, and threshold tuning.

3.5. Second improvement: MRI-anchored PET-optional fusion

The final model was created as MRI-anchored, PET-optional multimodal classifier. Two 3D encoders (MRI encoder and PET encoder) were used, each producing multi-scale feature maps. A multi-scale summary module performs global average pooling over intermediate and deep feature maps, concatenates resulting representations and passes the concatenation through a simple MLP with normalization and dropout.

PET representation can be considered as either trustworthy or not trustworthy. The optional PET is masked when PET information is missing or considered unreliable. Optional modality gate learns how much PET affects the final representation. Fused representation consists of concatenated MRI summary representation, optionally masked PET summary, and modality mask. In addition, auxiliary MRI/PET heads were introduced in order to reduce dependency on the main fusion path.

3.6. Training objectives and model selection

The training configuration employed class-balanced sampling and loss weighting for preventing majority-class collapse. For diagnosing AD, cross entropy or a class-balanced

Table 1. Results reported across the project notebooks. Values are percentages except MCC. The first-improvement row is a ds006148 OOD stress-test row, while the baseline and second-improvement rows are the controlled final-setting rows.

Notebook	Setting	Acc.	BAcc.	AUC	Sens.	Spec.	F1	MCC
Task 3 baseline	controlled	75.00	50.00	58.33	0.00	100.00	0.00	0.00
Task 4 first improvement	ds006148 stress test	65.75	50.00	44.00	0.00	100.00	0.00	
Task 5 second improvement	controlled	82.54	63.73	73.69	33.33	94.12	42.11	0.343

classification loss like focal were used according to the choice made in the notebook. Consistency losses helped the classifier make consistent predictions when PET was masked out. Model checkpoint selection did not consider only validation accuracy but non-collapse-aware metrics such as validation set AUC, F1, balanced accuracy, sensitivity, and MCC. Finally, an optimal decision threshold was determined for validation predictions and used for test prediction scoring.

The most relevant metrics are:

- Accuracy: percentage of correct predictions.
- Balanced accuracy: arithmetic mean of sensitivity and specificity.
- Sensitivity: recall of positive class, crucial for AD detection.
- Specificity: recall of negative class.
- AUC: quality of rank ordering across different thresholds.
- F1: harmonic mean of precision and sensitivity.
- MCC: correlation-based measure useful for imbalanced datasets.

4. Results

4.1. Main controlled split

Table 1 contains the summary for the main controlled split. The first improvement is reported here as an intermediary OOD test and not a controlled comparison within the same split because the associated notebook focused on the robustness to ds006148. In contrast, the controlled comparison here is between the baseline and the second-improvement experiment. It seems acceptable to judge the baseline based on its accuracy alone, but in reality, it has no positive predictions due to sensitivity and F1 values close to zero.

The final confusion matrix is:

$$\begin{bmatrix} 48 & 3 \\ 8 & 4 \end{bmatrix}$$

with the rows being the true classes and the columns being the predicted classes. There are 52 total correct predictions

among 63 test samples, achieving a test accuracy of 82.54%. However, more importantly, it stops predicting every single sample as belonging to the negative class. The sensitivity score remains below expectations, but at least now it is nonzero, and AUC increases to 73.69%.

4.2. ds006148 observations

It was very helpful to have some tests with ds006148 data to understand the limitations of the current approach. With the final controlled experiment design, we ended up with a total of 156 pairs of MRI and PET scans in our dataset, coming from 52 different subjects. The baseline model showed a class collapse problem, where the sensitivity score for the positive class is zero and the F1 score is also zero. Our first enhancement reduces the class collapse problem by increasing sensitivity to 33.33% and F1 to 42.11%.

5. Discussion

The key takeaway of the project is that more complex multimodal architecture is not necessarily better. Disease-acquisition factorization suggested by the survey on state-of-the-art was not applicable because the first step was to prevent collapse and leakage under low data. While the first improvement went towards more sophisticated ideas of harmonization and OOD, the existing pairs were insufficient for a large model. Meanwhile, the second improvement worked well because it returned to a simple idea – to preserve MRI as stable, train with a gate by PET, train with imbalance and select thresholds based on validation performance.

Besides the high accuracy of 82.54%, the main strength of the final model lies in its ability to improve AUC and recover the positive class, i.e., to distinguish AD cases in the dataset compared to a collapsed model. The last point is important since AD models can hardly be considered as valid if they achieve a good specificity at the expense of neglecting the positive class. However, the model fails to detect 8 cases, and thus suffers from a lack of sensitivity. At the same time, the model represents a true improvement over the baseline since it starts balancing the two classes instead of training exclusively on the majority class.

Finally, this project confirms an evaluation message from the literature. Indeed, without the sensitivity, F1, MCC, AUC, balanced accuracy, and confusion matrix, the collapse problem would remain unnoticed. This is exactly what AD studies need to address by applying leakage-aware splits and metrics.

6. Limitations

The first limitation lies in the small data size. Specifically, the controlled final split contains only 156 samples from 52

subjects. Moreover, the ds006148 paired dataset is limited, and, thus, it leads to unstable conclusions regarding robustness. Second, the implemented project cannot claim a full compliance with the idea suggested by SOA – a multimodal architecture explicitly implementing the latent factors of disease and acquisition, as well as residual leakage. Third, the positive class in question remains problematic since the final model improves sensitivity, but still misses several positive cases. Fourth, the experiments performed here cannot be considered as a SOA achievement due to resource limitations.

7. Conclusion

This paper started from a SOA research objective to develop invariant multiview AD prediction by factorizing disease-acquisition. The proposed pipeline for accomplishing this task was to create a baseline with an improvement I and II based on disease-acquisition factorization. The baseline presented the essential problem with the majority-class classifier collapse. Although accuracy was relatively decent, the balance between classes was skewed. The first improvement attempted at solving problems related to multimodal harmonization and reliability yet was unstable when applied on smaller ds006148 dataset. The second improvement delivered the most accurate model with a basic architecture of optional PET fused with MRI and class-balanced training/validation thresholding technique. On the final test set per subjects, it achieved 82.54% accuracy, 63.73% balanced accuracy, and 73.69% AUC. For future research, it is recommended to return to the initial objective by incorporating residual leakage detection modules and using a larger external paired cohorts datasets.

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